

PROVISIONAL PATENT APPLICATION

Phase-Completion Computing Devices: Winding-Number Register with Integer Store, Winding Arithmetic-Logic Unit with Modulus and Carry Control, Completion-Scheduled Commit Controller, Co-Signing Parametric Pair Gate, Condensate Collective Register, and Period-Finding Co-Processor, and Methods of Operating Them

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This specification is complete and self-contained. No companion application is relied upon; no material essential to written description, enablement, or the claims is incorporated from any other document. Every example under PROPHETIC EXAMPLES is prospective; no experiment, measured value, or nonzero performance result is represented as achieved.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

FIELD

The invention relates to information processing devices that encode and manipulate data in the accumulated phase and integer winding number of coherent oscillatory excitations in a physical medium, and more particularly to processors that alternate open-phase wave-interference stages with scheduled, irreversible state-commitment (readout-and-post) stages so as to hold durable quantized state in winding-number registers while performing arithmetic and period-finding operations. Applicable media include, without limitation, magnetostatic and exchange spin-wave (magnon) excitations in ferrimagnetic thin films (e.g., yttrium iron garnet), photonic and superconducting parametric oscillators, and acoustic/surface-acoustic-wave delay structures.

BACKGROUND

Two broad classes of non-conventional computing hardware are relevant background.

State-superposition machines. Gate-model quantum processors encode information in the superposed state of two-level systems and evolve that state unitarily. Preserving the manipulable state against uncontrolled environmental interaction is the dominant engineering burden. Superconducting and spin platforms require dilution refrigeration to approximately 10–15 mK, together with multi-stage magnetic, infrared, and vibration shielding and large numbers of attenuated control lines; ion and neutral-atom platforms substitute ultra-high-vacuum chambers and banks of frequency-stabilized lasers. Fault-tolerant operation further imposes an error-correction overhead: published surface-code estimates require on the order of 1,000–2,000 physical elements per protected logical element at practical code distances, i.e., roughly 10^3 – 10^4 physical elements per logical element across the useful-fidelity range, with the

overwhelming majority of physical elements existing solely to protect a small fraction of logical information (published surface-code and qLDPC estimates). The overhead grows as the protected state volume grows. Despite sustained investment, no such machine has demonstrated a commercially useful computation at scale (published field record).

Wave-interference computers. A separate lineage encodes bits in the phase of an oscillation (commonly 0 or π) and computes by interference: magnonic interferometer and majority-gate logic, oscillator- and optical-parametric Ising machines, parametron phase-bistable logic, and analog Fourier/interference processors. These machines are durable against the cryogenic burden above but lack (a) a durable, quantized, multi-valued state element and (b) a scalable arithmetic architecture. Interference approaches to factoring/period-finding based on truncated Gauss sums suffer a resource scaling in which the number of summation terms required to suppress false ("ghost") resonances grows as a power of the target integer, so resources scale exponentially in the number of digits and no polynomial speedup is obtained (the published Gauss-sum ghost-factor analyses). Such phase-bistable and interference schemes provide no whole-number quantized register, no scheduled irreversible commitment of intermediate results, and no arithmetic in an integer state space.

Single-flux-quantum logic. Rapid single-flux-quantum (RSFQ/SFQ) superconducting logic uses a 2π Josephson phase slip as a voltage pulse and stores integer flux quanta in loops. It is a pulse-logic family: it provides no open-phase interference stage between commits, no scheduled commit acting as an error-digitizer against continuous drift, no multi-radix or residue registers with radix conversion, and no parametric pair source or condensate bus.

The gap. There remains a need for a computing architecture that combines (i) a durable, quantized, multi-valued state element — an integer winding-number register whose integer is held in a disclosed integer store and is robust by whole-turn quantization at commit — with (ii) fast, cheap, but perishable open-phase interference used only inside bounded-depth stages, and (iii) scheduled, deliberate irreversible commit operations that digitize interference results into the winding registers rather than attempting to protect an open state indefinitely. The present invention supplies that architecture and its constituent apparatus and methods.

Applicant's published disclosure (nonessential background). Applicant published, on 2026-08-30, a public exhibit presenting the architecture's concepts in didactic depth ("Computation as Completion," <https://cfmilam.github.io/nr-poams-exhibits/computation-as-completion.html>), part of applicant's commit-dated exhibit series. That page is applicant's own prior publication and is cited as nonessential background only: no material essential to enablement, written description, or intended claim scope is incorporated from it, and this specification is complete and self-contained.

SUMMARY OF THE INVENTION

Disclosed are apparatus and methods for phase-completion computing. Data is held durably as the integer winding number (the accumulated whole-turn count) of a coherent oscillatory excitation, kept in an integer store of a disclosed species, and is manipulated by alternating two kinds of stage: an open-phase

interference stage, in which continuous phase evolves and combines linearly in a wave medium, and a commit stage, in which the accumulated phase is irreversibly read out against a reference and posted to a winding-number register as a quantized whole-turn transition. The commit stage functions as an error-digitizer: it collapses continuous phase drift onto the nearest whole turn, so sub-turn error is quantized at each commit rather than accumulated across the computation. This inverts the design premise of state-superposition machines, which forbid such completions and pay to protect an open state; here, completions are scheduled and used as the primitive that produces durable state.

The products and their operating methods, corresponding to five families, are:

- **H1 — Winding register and winding ALU:** a winding-number register comprising a medium element holding the open (fractional) phase and an integer store of one of three disclosed species — a commit counter, a residue (multi-radix) register bank, or a topological medium state — and an arithmetic-logic unit performing addition, subtraction, and multiplication in winding space, with a per-register modulus, carry and borrow implemented as quantized whole-turn transition events at the modulus boundary, a mode control selecting positional or modular arithmetic, and radix conversion between registers of differing per-turn quanta.
- **H2 — Completion-scheduled commit controller and processor:** an architecture and clocking method that alternates open-phase interference stages with irreversible commit stages, sets the commit period from the measured error distribution and the finite computation-length budget disclosed below, and uses the commit as an error-digitizer. The architecture confines sub-turn drift between commits; whole-turn miscounts remain and are budgeted rather than presumed absent.
- **H3 — Co-signing gate:** deterministic generation of shared/correlated circulation between two or more oscillatory modes by parametric pair creation, providing a two-seat (correlated-pair) logic primitive whose commits at the two seats are related through a pump-locked phase sum. ("Co-signing" is defined herein as a controlled operation that places a single quantity of shared angular momentum into a definite correlated relation across two or more modes such that a subsequent commit at either mode is deterministically related to the other; it is used as an engineering term of art only.)
- **H4 — Condensate collective register:** a room-temperature magnon-condensate phase register, written by a phase-referenced seed, read by a matched pickup and commit, and steered by bias field, serving additionally as a collective interference bus.
- **H5 — Period-finding engine:** a co-processor combining the winding ALU, bounded-depth interference, and commit readout to perform order-finding/period-finding, in a first operating mode computing the modular power sequence in winding hardware by the disclosed multiplication path and in a second operating mode accepting host-supplied modular powers, packaged as a co-processor to a conventional host. No asymptotic speedup over classical order-finding is asserted.

The architecture confines continuous-phase precision demands to bounded-depth interference stages between commits and keeps all persistent arithmetic in quantized winding space, so as to mitigate the precision-scaling limitation that constrains prior interference-only period-finding schemes.

BRIEF DESCRIPTION OF THE DRAWINGS

(Eleven sheets; 1000-series numerals, one thousand-series per sheet.)

- **FIG. 1000** — System overview: winding registers (1010), interference medium (1020), co-signing source (1030), condensate bus (1040), commit subsystem (1050), host interface (1060), and controller/scheduler (1070), showing the load → evolve → commit → post loop.
- **FIG. 2000** — H1 winding ALU: integer-turn accumulator (2010) with whole-turn counter (2020); carry element (2030) emitting a quantized transition to a higher-order register (2040) at the modulus boundary; radix-conversion coupling (2050, 2052, 2054) between registers of differing per-turn quanta.
- **FIG. 3000** — H2 completion-scheduled timing diagram: stage clock (3010), open-phase stages (3020) and commit stages (3030); accumulated phase (3040) resolving to the nearest whole turn at each commit; commit-as-error-digitizer (3050); half-turn discrimination bound and commit-schedule relation (3060).
- **FIG. 4000** — H3 co-signing gate: parametric pump (4010) driving correlated mode-pair creation between seats (4020a, 4020b); shared correlated relation (4030); related commits (4040); stand-alone correlation-source variant (4050); phase-sum commissioning control and prophetic CHSH discriminator.
- **FIG. 5000** — H4 condensate collective register: ferrimagnetic film (5010), parametric pump (5020), bias-field steer (5030), condensate collective phase (5040), commit reader (5050) with matched pickup, many-to-one interference bus (5060), phase-referenced seed antenna (5070), and pickup antenna (5080).
- **FIG. 6000** — H5 period-finding engine: winding ALU (6010), bounded-depth interference stage (6020), commit readout (6030), host interface (6040), periodicity extractor (6050); committed winding series showing periodicity at $k = r, 2r, 3r$.
- **FIG. 7000** — Alternative-media embodiments (schematic): magnonic (7010), photonic (7020), and superconducting-parametric or acoustic (7030) realizations of the same architecture, each with its named integer-store element.
- **FIG. 8000** — Integer-store species: (A) commit-counter register (8010) with reference oscillator (8012), phase comparator (8014), wrap detector (8016), and up/down counter (8018); (B) residue register bank (8020) with coprime moduli, per-channel quantum $2\pi/m_i$, and CRT recombination (8022); (C) topological medium-state register (8030) — fluxoid-number loop (8032), ring-mode index (8034), condensate pattern position (8036) — labeled prophetic.

- **FIG. 9000** — Modulus, carry, and borrow control: register with modulus M (9010), modulus-boundary detector (9020), carry/borrow pulse generator (9030), mode control input (9040) selecting positional or modular arithmetic, higher-order register (9050); worked example $M = 8$: $7 + 1 \rightarrow 0$ with carry 1; $0 - 1 \rightarrow 7$ with borrow 1.
- **FIG. 10000** — Multiplication path and engine operating modes: multiplicand register (10010), multiplier control sequence (10020), doubling (shift) step (10030), conditional modular addition (10040), product register (10050); mode A hardware modular exponentiation by iterated multiplication (10060); mode B host-supplied modular powers (10070).
- **FIG. 11000** — Layered scope and qualification: layers L0 raw interference observable (11010) $\rightarrow$ L1 committed integer (11020) $\rightarrow$ L2 arithmetic operation (11030) $\rightarrow$ L3 application (11040) with validity gates between layers; commissioning benchmarks B1–B5 (11050); controller validity-state machine — qualify, valid, degraded, invalid, safe state (11060).

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels appearing in quotation marks identify phenomena by their customary names and commit neither the specification nor the claims to any entity ontology historically associated with those names. Every claimed quantity is fixed by a recited element or reading. Hypothesis statements are identified expressly and are not relied on for utility.

1. **Phase (open phase)** — the instantaneous angular argument, in radians, of a coherent oscillatory excitation in the medium relative to a reference oscillator; "open" means not yet committed. An interference output depends on phase modulo 2π only.
2. **Winding number** — the integer count of whole turns (2π accumulations) of the phase relative to the reference oscillator, held as a persistent integer in an integer store.
3. **Integer store; register species** — the element holding the winding number. Three species are disclosed: (A) a **commit-counter register**, in which an electronic or other digital up/down counter is incremented or decremented at each committed wrap of the open phase and the medium holds only the open phase; (B) a **residue register bank**, in which several channels of coprime moduli $m_1, m_2, \dots$ each hold a residue r_i as a phase $r_i \cdot 2\pi/m_i$ and the integer is recovered by Chinese-remainder (CRT) recombination; (C) a **topological medium-state register**, in which the medium itself holds an integer as a quantized state (fluxoid number of a superconducting loop, mode index of a ring resonator, or period-position of a two-component condensate pattern). Species C is prophetic in this specification.

4. **Winding-number register** — a medium element holding open phase together with an integer store of a disclosed species and a commit detector.
5. **Commit (completion, closure)** — an irreversible operation that reads the open phase against the reference oscillator, resolves it to the nearest whole turn (or nearest quantum for species B), updates the integer store accordingly, and discards the sub-turn residual; a commit is not reversed within the computation.
6. **Open-phase stage; interference stage** — an interval during which phase evolves and combines continuously in the medium without commitment; the element in which two or more open phases are combined.
7. **Completion schedule; commit period T** — the controller's timing rule fixing the interval T between commits and the number D of combining operations (interference depth) permitted between commits.
8. **Commit-schedule relation** — for the disclosed zero-mean Gaussian independent-increment model, $p_{\text{miss}}(T) = \text{erfc}(\pi/(\sigma\sqrt{2T}))$ and the exact union-bound condition is $C p_{\text{miss}}(T) \leq \varepsilon$, equivalently $T \leq \pi^2 / \{2\sigma^2 [\text{erfc}^{-1}(\varepsilon/C)]^2\}$. For measured non-Gaussian or correlated drift, the controller uses the measured tail distribution or a conservative confidence bound. Here σ^2 is phase-drift variance per unit time, C is computation length in commits, and ε is the target total miscount budget.
9. **Discrimination bound** — the largest open-phase excursion resolvable to the correct value at commit: one half-turn (π) for species A and C, one half-quantum (π/m_i) for channel i of species B.
10. **Guard band δ** — a phase interval about the decision boundary within which the commit comparator re-samples or flags the value rather than resolving it.
11. **Modulus M** — the whole number of turns at which a register wraps to zero.
12. **Carry; borrow** — a single quantized whole-turn transition emitted to the next higher-order register when a commit carries a register across its M -th turn upward (carry) or across zero downward (borrow).
13. **Mode control** — a control input selecting positional arithmetic (carry and borrow enabled; base- M digits) or modular arithmetic (carry and borrow disabled; arithmetic modulo M).
14. **Radix; per-turn quantum** — the angular scale ($2\pi/m$) of one register unit; radix conversion re-expresses a count in one register as a count in a register of different per-turn quantum.
15. **Multiplication path** — the control sequence of doubling (shift) steps and conditional modular additions by which a product of two register values is formed from the addition, wrap, and carry primitives; or residue-wise multiplication in species B.
16. **Co-signing gate** — a parametric pump driven at approximately twice a mode frequency creating correlated mode pairs $(k, -k)$ whose phase sum $\phi_A + \phi_B$ is locked to the pump phase, with a commit detector at each seat.
17. **Condensate register** — a magnon condensate in a ferrimagnetic film maintained under parametric pumping, its collective phase read against a reference and held by an integer store.

18. **Period-finding engine** — a co-processor comprising a winding ALU, a bounded-depth interference stage, a commit readout, a periodicity extractor, and a host interface, operable in a hardware-power mode or a host-supplied-power mode.
19. **Reference oscillator** — the stable oscillator against which open phase is measured at commit; may be a dedicated reference register in the same medium (differential read).
20. **Validity states** — *qualify* (commissioning in progress), *valid* (all gates pass; L2 results reported), *degraded* (a runtime witness out of tolerance; results flagged), *invalid* (a gate failed; no L2 result reported), and **safe state** (commits suspended, integer stores frozen and read-only, pump off).
21. **Commissioning benchmark; acceptance gate** — a measured quantity obtained during qualification, and the threshold it must satisfy before the corresponding layer may be reported.

DETAILED DESCRIPTION

Terminology. Terms are used as defined in DEFINITIONS. "Co-signing" is defined in the Summary and DEFINITIONS. These terms are engineering terms of art in this specification and carry no interpretation beyond the operations defined.

Overview of operation (FIG. 1000). A phase-completion processor (1000) holds durable data in one or more winding-number registers (1010). A computation proceeds as a scheduled sequence: (a) load registers from the host interface (1060); (b) drive one or more bounded-depth open-phase interference stages in the interference medium (1020), in which register contents are expressed as phases and combined; (c) execute a commit stage in the commit subsystem (1050) that digitizes the interference result back to integer winding values and posts them to the integer stores; (d) repeat under the controller/scheduler (1070). A co-signing source (1030) and a condensate bus (1040) are optional resources. Continuous phase is treated as a fast, cheap, perishable compute fluid; winding integers in the integer stores are the durable state.

LAYERED SCOPE (FIG. 11000)

L0 — raw interference observable (11010). The phase-sensitive detector output at commit: an in-phase/quadrature pair or a comparator threshold crossing referenced to the reference oscillator, giving the open phase modulo 2π with its measured uncertainty. L0 is always reportable. **L1 — committed integer (11020).** The integer-store value after a commit: the wrap count (species A), the residue vector and its CRT recombination (species B), or the read quantized state (species C). Reportable when the open-phase excursion since the previous commit lies within the discrimination bound less the guard band and the commit-schedule relation is satisfied for the measured σ^2 . **L2 — arithmetic operation (11030).** A sum, difference, product, carry/borrow chain, or radix conversion posted to destination registers. Reportable when every contributing commit is L1-valid and the controller is in the *valid* state; otherwise the operation is flagged *degraded* or withheld *invalid*. **L3 — application (11040).** Order-finding results, correlated-pair resources, condensate-bus combining, and host-visible outputs. Reportable only from L2-valid operations. No statement of computational speedup is made at any layer.

QUALIFICATION, COMMISSIONING, AND CALIBRATION (FIG. 11000)

Factory qualification vs. deployed operation. Each unit or production class is qualified once with the full benchmark set B1–B5 and a stored drift model; deployed operation uses the stored commit period, guard band, and reference assignments with a reduced set of runtime witnesses. A deployed unit never reports an L2 result its qualification does not support. **Commissioning benchmarks (11050).** *B1 drift variance:* with commits disabled, record open phase against the reference oscillator over many intervals; fit σ^2 and any systematic offset $\Delta\omega$; store both. *B2 commit fidelity:* run a known load-add-commit sequence at commit periods $2T$, T , $T/2$ and confirm the miscount rate follows $p_{\text{miss}} = \text{erfc}(\pi/(\sigma\sqrt{2T}))$ with the measured σ^2 and that the offset term $|\Delta\omega|T$ plus the tail remains within the half-turn budget; record the boundary-proximity histogram and set the guard band δ . *B3 carry/borrow:* over many trials confirm the higher-order register equals $\lfloor \text{total}/M \rfloor$ against the host and that no double count occurs on re-crossings. *B4 pair source (H3):* vary the pump phase and confirm the two-seat phase sum tracks it with variance below threshold while each single-seat phase is uniformly distributed over trials; confirm the relation survives a pump-off delay before readout and is absent when one seat is blocked (crosstalk control). *B5 condensate register (H4):* confirm that the read phase is at ω_{min} and not at the pumped $\omega_{\text{p}}/2$ by frequency discrimination and by read after a pump-off delay; confirm the steering law $\Delta\Phi = \gamma\Delta H \cdot \tau$ by varying bias step and duration. **Acceptance gates, recalibration, and fail-safe.** Gate on B1–B5 with preregistered thresholds; store σ^2 , $\Delta\omega$, δ , T , M , mode assignments, and reference-register assignments in a calibration memory with provenance. Runtime witnesses are the reference-register phase, the boundary-proximity count, and the miscount count from redundant copies; a witness out of tolerance moves the controller to *degraded* and shortens T ; a failed gate moves it to *invalid* and triggers recalibration of σ^2 ; loss of the reference oscillator or pump moves it to the **safe state** — commits suspended, integer stores frozen and read-only, pump off, host notified.

H1 — Winding Register and Winding ALU (FIGS. 2000, 8000, 9000, 10000)

A winding-number register comprises a medium element whose open phase is the working variable, an integer store of a disclosed species holding the winding number, a reference oscillator, and a commit detector. In a magnonic embodiment the medium element is a resonant or delay-line structure whose round-trip phase, tracked modulo 2π against the reference, carries the open phase; the integer store (2020) holds the register value. The interference output of the medium carries the phase modulo 2π only; the integer is held by the integer store and is updated at commit as follows for each species.

Addition and subtraction. Addition is performed by expressing the addends' open phases in the interference stage, combining them, and committing: the commit detector reads the combined phase against the reference, the wrap detector counts the number of whole turns crossed since the previous commit, and the integer store is incremented by that count and by the integer parts of the addends carried from their stores. Subtraction is performed by combining a phase increment of opposite sense, decrementing the store. Because turn count is inherently periodic, arithmetic modulo a chosen modulus is

realized directly by choosing the register's per-turn quantum and modulus M so that the modulus corresponds to a whole number of turns; with the mode control set to modular, the register wraps at M without an explicit modulo operation.

the modulus, carry, and borrow subsection Modulus, carry, and borrow (FIG. 9000). Each register (9010) has a modulus M , a whole number of turns. When a commit carries the register across its M -th turn upward, the modulus-boundary detector (9020) causes the carry/borrow generator (9030) to emit exactly one carry — a single quantized whole-turn transition — to the adjacent higher-order register (9050, also 2040 in FIG. 2000), and the low register wraps to zero; when a commit carries the register across zero downward, one borrow is emitted and the low register wraps to $M - 1$. Carry and borrow are thus physical, quantized, discrete events, detectable and clocked, and are emitted only at the modulus boundary, not at every whole turn. A mode control input (9040) selects positional arithmetic (carry and borrow enabled, base- M digits chained across registers) or modular arithmetic (carry and borrow disabled, arithmetic modulo M in a single register). Worked example ($M = 8$): low register 7, add 1 $\rightarrow$ low register 0, carry 1 to the higher-order register; low register 0, subtract 1 $\rightarrow$ low register 7, borrow 1.

Radix conversion (FIG. 2000, 2050–2054). Radix conversion between registers of differing per-turn quanta (differing angular scale or wavelength) is performed in two disclosed cases. (i) Integer-ratio conversion: where the source and destination quanta stand in an integer ratio q , a count of N turns in the source is expressed by coupling the two registers through an interference stage and committing, yielding $N \cdot q$ (or $\lfloor N/q \rfloor$ with a remainder register) turns in the destination. (ii) Residue-to-positional conversion (species B to species A): the integer X is recovered as the weighted sum $X = \sum r_i \cdot w_i \bmod M$, with $M = \prod m_i$ and $w_i = (M/m_i) \cdot [(M/m_i)^{-1} \bmod m_i]$, executed as constant-multiply-and-accumulate by the multiplication path disclosed in the multiplication-path subsection (repeated modular addition, or phase multiplication by k -th harmonic generation followed by commit). Conversion between quanta of non-integer ratio is approximate to the nearest whole turn and is so flagged.

Species A — commit-counter register (FIG. 8000, 8010). The medium holds open phase modulo 2π . A phase comparator (8014) compares the medium phase with the reference oscillator (8012); a wrap detector (8016) registers each crossing of the 2π boundary in either sense during the open stage; an up/down counter (8018) is incremented or decremented at each committed wrap. The integer lives in the counter; the interference stage handles only the open (fractional) phase. The open phase is sampled at not fewer than two samples per half-turn of relative phase excursion so that no wrap is missed, and the commit comparator applies the guard band δ disclosed in the guard-band subsection. A commit recovers an integer by (a) counting the wraps since the previous commit, (b) rounding the residual open phase to the nearest whole turn, and (c) adding (a) and (b) to the counter. Species A is buildable from disclosed elements with ordinary skill.

Species B — residue register bank (FIG. 8000, 8020). A bank of channels with pairwise coprime moduli $m_1, m_2, \dots, m_j$ (for example 7, 8, 9; $M = 504$) holds the value X as residues $r_i = X \bmod m_i$, each channel storing r_i as the phase $r_i \cdot 2\pi/m_i$. Interference adds residues natively modulo m_i ; a commit in channel i snaps the phase to the nearest multiple of $2\pi/m_i$, so the discrimination bound for that channel is one half-

quantum π/m_i , and the commit-schedule relation disclosed in the commit-schedule subsection applies with π replaced by π/m_i (T shrinks by a factor m_i^2). A commit recovers the integer by reading each channel's residue and recombining by CRT (8022) by the residue-to-positional conversion disclosed above. Three coprime channels tie to the $r = 3$ redundancy disclosed in the commit-schedule subsection: the same three channels supply an integer-majority check on the recombined value. Worked example: $X = 123$; residues $(4, 3, 6) \bmod (7, 8, 9)$; weights $(288, 441, 280)$; $\sum r_i w_i \bmod 504 = 123$.

the topological medium-state register subsection Species C — topological medium-state register (FIG. 8000, 8030) — prophetic. The medium itself holds the integer as a quantized state: (a) the fluxoid number of a superconducting loop (8032), a Josephson phase slip being the quantized whole-turn transition and thus a physical carry; (b) the mode index of a ring resonator (8034); (c) the period-position of the standing-wave density pattern of a two-component ($\pm k_{\min}$) magnon condensate (8036), displacement of the pattern by one period being one turn. A commit in species C reads the quantized state directly (flux readout, mode-index readout, or pattern-position readout) and posts it to a shadow counter for host access. No species-C register has been built by the applicant; the species is disclosed for scope with the structures named.

the multiplication-path subsection Multiplication path (FIG. 10000). Multiplication of two register values is performed from the addition, wrap, and carry primitives under a control sequence: the multiplicand register (10010) is doubled by adding it to itself (shift step, 10030) once per bit of the multiplier; the multiplier control sequence (10020), read from the multiplier register or supplied by the controller, conditionally adds the current doubled value to the product register (10050) modulo M (conditional modular addition, 10040) when the corresponding multiplier bit is one. For a multiplier of b bits the product requires at most b doublings and b conditional additions, each followed by a commit.

Multiplication by a constant a is alternatively performed by a -fold repeated modular addition. In species B, multiplication is performed residue-wise: each channel multiplies modulo m_i by the same path.

Multiplication by an integer constant k is alternatively performed by k -th harmonic generation of the open phase followed by commit, where the medium supports harmonic generation.

Redundancy and reference. In embodiments each register is triplicated ($r = 3$) and the committed integers are majority-voted; because field noise on one chip is common-mode, the primary drift control is a differential read against a reference register in the same medium (species A with the reference register serving as reference oscillator), majority voting being secondary.

H2 — Completion-Scheduled Commit Controller and Processor (FIG. 3000)

A completion-scheduled processor comprises a control subsystem (1070) that clocks a repeating alternation of at least one open-phase interference stage (3020) and at least one commit stage (3030), and a set of winding-number registers written by the commit stages. During an open-phase stage the control subsystem enables continuous phase evolution and interference among selected modes; during a commit stage it triggers irreversible readout that resolves accumulated phase to whole-turn values against the reference and posts them to the integer stores.

The commit stage operates as an error-digitizer (3050). Continuous phase drift accumulated during an open-phase stage is resolved, at commit, to the nearest whole turn, so that sub-turn error present at the moment of commit is discarded rather than propagated. Sub-turn error therefore does not accumulate across successive stages; the probability of a whole-turn miscount accumulates linearly with the number of commits and is budgeted by the relation disclosed in the commit-schedule subsection. The scheduling method comprises choosing the interval between commits so that worst-case accumulated drift, including any systematic offset, remains within the discrimination bound less the guard band.

the clocking and scheduling subsection Clocking and scheduling. The control subsystem defines a commit period and a stage schedule; in embodiments the commit period is derived from the medium's oscillation frequency (for magnonic media, in the approximately 1–100 GHz range per the published field record) and the interference-stage depth is bounded to a fixed number of combining operations between commits. The schedule may be static or adaptively shortened when measured drift approaches the discrimination bound.

Positioning relative to error correction. The disclosed processor confines analog phase precision to the interval between commits and holds persistent results in an integer store. This can substitute scheduled measurement, guard bands, retry, redundancy, and safe-state operation for some forms of continuous protected-state overhead. It does not prove that error-control overhead is constant or absent: whole-turn miscounts accumulate with computation length, correlated drift can defeat the independent model, and a fixed total error budget can require a shorter commit interval, increased redundancy, checkpoint/restart, or another qualified control as C increases. Maintenance of a pumped medium (H4) is separately measured and may scale with medium size, mode count, duty cycle, and implementation.

the commit-schedule subsection Commit-schedule relation (3060). For zero-mean Gaussian independent phase increments with measured variance rate σ^2 , the per-commit whole-turn miscount probability is $p_{\text{miss}}(T) = \text{erfc}(\pi/(\sigma\sqrt{2T}))$. For C commits and target total budget ϵ , the union bound requires $C p_{\text{miss}}(T) \leq \epsilon$; the exact Gaussian design limit is therefore $T \leq \pi^2 / \{2\sigma^2 [\text{erfc}^{-1}(\epsilon/C)]^2\}$. The simpler logarithmic expression $\pi^2 / [2\sigma^2 \ln(C/\epsilon)]$ is only a labeled approximation whose coverage must be checked against the exact tail. With three independent replicas, majority error per commit is $3p_{\text{miss}}^2(1-p_{\text{miss}})+p_{\text{miss}}^3$; total failure is bounded by C times that quantity. Fixed redundancy does not make total error independent of C . The controller selects T , redundancy, retry/checkpoint policy, and guard band from the measured distribution and target budget. A systematic offset $\Delta\omega$ adds a deterministic term so that $|\Delta\omega|T + \text{tail} \leq \pi - \delta$; correlated or non-Gaussian residuals require an empirical tail bound. For species B, π is replaced by π/m_i . Interference depth is bounded by the operations executable inside the qualified T ; no universal $D \propto 1/\ln C$ law is asserted.

Guard band and metastability. A phase sitting at the decision boundary at commit has undefined rounding. The commit comparator applies a guard band $\delta < \pi$ about the boundary: a value within δ of the boundary is re-sampled after a short interval, or flagged and the commit repeated at the next stage clock; the boundary-proximity count is a runtime witness (described under acceptance gates and fail-safe operation).

H3 — Co-Signing Gate (FIG. 4000)

A co-signing gate deterministically generates shared or correlated circulation between two or more oscillatory modes. In a magnonic embodiment the gate comprises a parametric pump (4010) that, driven at approximately twice a mode frequency (parallel pumping), creates correlated mode pairs (k , $-k$) at seats (4020a, 4020b), placing a defined quantity of shared angular momentum into a correlated relation across the two seats (4030): the phase sum $\phi_A + \phi_B$ is locked to the pump phase plus a constant, while each single-seat phase is undetermined unless seeded. A subsequent commit at either mode is therefore deterministically related to the commit at the other through the pump-referenced phase sum (4040). Each pair carries wave-vectors k and $-k$ and removes two quanta of the pumped mode's angular momentum projection.

The gate provides a two-seat logic primitive: operations are defined over the correlated pair as a unit, so that a single controlled circulation supports paired conditional operations without independent preparation of each seat. The correlated relation persists for the mode coherence time and is fixed at commit; the committed values are then held in the integer stores.

A correlation source embodiment (4050) provides the co-signing gate as a stand-alone generator of correlated mode pairs on demand, with control inputs selecting the participating modes (by bias field and geometry) and the sense of the shared circulation (by pump phase), and a pump-power control held above the parametric threshold, for use as a resource by the processor of H2 and the engine of H5.

Correlated-pair source commissioning benchmark (B4). Its first acceptance gate is the phase-sum benchmark: over repeated trials, the variance of the two-seat phase sum $\phi_A + \phi_B$ lies below a preregistered threshold and the sum tracks a commanded change of pump phase, while each single-seat phase is uniformly distributed over trials; the relation survives a pump-off delay before readout and vanishes when one seat is blocked. A gate meeting this benchmark is performing to specification as a correlation resource. This control establishes source operation without deciding the ontology of the shared two-seat exchange.

POAMS-native co-signing discriminator (prophetic; not represented as achieved). The two seats are treated as the two ends of one completed exchange rather than independent local variables. With four preregistered analyzer settings and binary committed outcomes, the predicted correlation follows the co-signed phase relation and approaches the CHSH magnitude $2\sqrt{2}$ in the low-thermal-occupation, high-visibility limit. The experiment shall report the raw four correlation tables, singles, coincidence/commit windows, setting selection, loss and no-commit events, and the computed CHSH statistic without fair-sampling deletion. Failure to exceed 2 under the preregistered validity envelope falsifies this prophetic co-signing branch; it does not invalidate the phase-sum gate or the remaining computing architecture. The local classical $|S| \leq 2$ result is retained as the explicit competing control, not assumed as the ontology of the device.

H4 — Condensate Collective Register (FIG. 5000)

A condensate collective register comprises a magnon condensate in a ferrimagnetic film (5010) maintained under parametric pumping (5020), the collective coherent phase Φ (5040) of the condensate serving as the open phase of a register whose integer store is of species A (or, prophetically, species C(c)). Room-temperature magnon condensation under parametric pumping in yttrium iron garnet films is an established phenomenon in the field record (Demokritov et al., Nature 443, 430–433 (2006)); the condensate forms at the spectral minimum $\omega_{\min}$ ($k_{\min} \neq 0$), below the pumped frequency $\omega_p/2$, and its phase is spontaneously chosen unless seeded. The present register uses the collective phase as a macroscopic, coherently addressable state variable.

Write, read, and steer. *Write*: a phase-referenced seed signal at $\omega_{\min}$, derived from the reference oscillator, is injected through a $k_{\min}$ -matched seed antenna (5070) during condensate formation, locking Φ to the seed phase; alternatively the register is written by bias-step steering from a previously read state. *Steer*: a bias-field step ΔH (5030) for a duration τ shifts $\omega_{\min}$ by $\gamma\Delta H$ and advances Φ by $\Delta\Phi = \gamma\Delta H \cdot \tau$ relative to the reference; in embodiments the bias-field and steering control comprise programmable multi-axis field-positioning hardware. *Read*: a $k_{\min}$ -matched inductive pickup antenna (5080) (or phase-resolved Brillouin light scattering) is mixed against the reference oscillator at $\omega_{\min}$, and the commit reader (5050) resolves Φ to a whole turn with the wrap counter of species A, after a pump-off delay that discriminates the condensate at $\omega_{\min}$ from pumped magnons at $\omega_p/2$.

Interference bus (5060). Because the condensate is a single collective coherent object, it serves as an interference bus onto which multiple mode contributions are combined by $k_{\min}$ -matched coupling antennas for a shared open-phase stage prior to a common commit, providing a many-to-one combining element for the processor of H2. Bus operation is prophetic.

Pattern-position species (prophetic). A two-component ($\pm k_{\min}$) condensate forms a spatial standing-wave density pattern whose position is the relative phase of the two components; displacement of the pattern by one period is one turn, and the pattern position is a directly observable, winding-capable variable read by spatially resolved Brillouin light scattering — species C(c) described under the topological medium-state register.

H5 — Period-Finding Engine (FIG. 6000, FIG. 10000)

A period-finding engine comprises the winding ALU of H1 (6010), at least one bounded-depth interference stage (6020), a commit-readout stage of H2 (6030), a periodicity extractor (6050), and a host interface (6040), arranged to perform order-finding. Register contents representing successive powers $a^k \bmod N$ of a target integer a under a chosen modulus N are expressed as phases, combined in the bounded interference stage, and committed; periodicity in the committed winding values — the first $k = r$ at which the committed value returns to $a^0 = 1$, and its repeats at $2r, 3r$ — reveals the order sought.

The engine confines continuous-phase precision demands to the bounded-depth interference stages and keeps all persistent arithmetic in quantized winding registers, and is thereby directed to mitigating the accumulated-precision growth that limits interference-only (Gauss-sum-class) period-finding (the published Gauss-sum analyses); sub-turn precision demands are not carried across commit boundaries.

the co-processor-packaging subsection Co-processor packaging. The engine is packaged as a co-processor to a conventional digital host, exposing register load/store and a commit-triggered result readout over a host interface, the host supplying the modulus and target integer and receiving committed winding values for classical post-processing.

Scope. The engine is a classical-complexity-class order-finding co-processor: no asymptotic speedup over classical order-finding is asserted; the engine steps through at least r values. The engine's merit is native modular arithmetic in winding space, commit-disciplined error, and throughput and energy economics — in the winding representation the turns are free and the postings are tolled: the energy budget attaches to commit operations, not to open-phase evolution. This energy statement is an intended effect: in damped media the open phase must be re-excited and a pumped medium (H4) draws maintenance power, which enters the budget as a per-medium term.

Operating modes (FIG. 10000). *Mode A — hardware powers (10060)*: the engine computes $a^{k+1} \bmod N$ from $a^k \bmod N$ by one modular multiplication per step using the multiplication path disclosed in the multiplication-path subsection, commits, and posts to the register series. *Mode B — host-supplied powers (10070)*: the host computes $a^k \bmod N$ and loads each value; the engine performs modular accumulation, commit, and periodicity extraction in winding hardware. In mode B the engine's winding hardware is a modular register memory with committed readout and a periodicity detector, and is so described; the ALU label of FIG. 6000 applies to mode A.

EMBODIMENTS

Primary embodiment — magnonic / YIG. The primary embodiment realizes the registers, gates, condensate register, and interference stages in yttrium iron garnet (YIG) thin films and waveguides, using spin-wave (magnon) excitations at frequencies in the field-recorded range (approximately 1–100 GHz, higher for exchange/antiferromagnetic modes), phase-encoded interference in waveguide structures, parametric pumping for co-signing pair creation and for condensate maintenance, bias-field steering, and a species-A integer store in classical control logic. YIG is selected for its low magnetic damping. No specific device dimensions, damping figures, or fidelity values are asserted here beyond the qualitative field record; concrete parameters are to be supplied from bench measurement under the qualification procedure of the layered-scope and qualification subsections.

the alternative-media embodiments Alternative media (FIG. 7000; each prophetic; named for scope). The same architecture — winding registers, scheduled commits, co-signing pair sources, collective interference bus, and period-finding stack — may be embodied in other coherent oscillatory media, including: (a) photonic media (7020), e.g., degenerate optical parametric oscillators and interferometer meshes, wherein the open phase is carried by optical fields, co-signing by optical parametric down-

conversion, and the integer store is a species-A counter against a local oscillator or, as species C, the mode index or orbital-angular-momentum charge of a ring resonator; (b) superconducting-parametric media (7030), e.g., Josephson/Kerr parametric oscillators, wherein the open phase is carried by resonator states, co-signing by parametric pumping, and the integer store is the fluxoid number of a superconducting loop with a Josephson phase slip as the quantized whole-turn transition (species C) or a species-A counter; and (c) acoustic media (7030), e.g., surface-acoustic-wave and delay-line structures, wherein the open phase and interference are carried by acoustic excitations and the integer store is a species-A counter. The listing is illustrative and not limiting.

Hybrid and control embodiments. The processor may be operated as a co-processor to a conventional digital host (H5 the co-processor-packaging subsection); the commit subsystem may use threshold or phase-sensitive (I/Q) detection against the reference oscillator to resolve whole-turn values; and the scheduler may be implemented in classical control logic that adaptively sets the commit period under the clocking and commit-schedule subsections. Arrays of registers share one medium, one reference oscillator, and one scheduler; carry chains of more than two registers are formed by repeating the carry/borrow element of the modulus, carry, and borrow subsection between adjacent registers.

Prophetic labeling. Embodiments and performance statements not accompanied by measured values are prophetic. Species A registers, the ALU, the commit controller, the classical pair source, and mode-B period-finding are buildable from the disclosed elements with ordinary skill; species B is buildable from species-A elements with a channel bank; species C, the condensate bus, the pattern-position register, alternative media, and the energy-economics statement are prophetic.

PROPHETIC EXAMPLES (prospective; every number follows from the stated parameters; no result is represented as achieved)

P1 — commit schedule. Assume a measured Gaussian independent-increment model with $\sigma^2 = 0.01 \text{ rad}^2/\text{ns}$ (to be measured in B1), $C = 10^6$ commits, and $\varepsilon = 10^{-3}$. The controller solves $C \cdot \text{erfc}(\pi/\sqrt{(2\sigma^2 T)}) \leq \varepsilon$ numerically or through inverse erfc. Choosing $T = 20 \text{ ns}$ gives $p_{\text{miss}} = \text{erfc}(4.97) \approx 2.1 \times 10^{-12}$ and $C p_{\text{miss}} \approx 2.1 \times 10^{-6}$, inside the budget. With three independent replicas, the majority-error approximation is $3C p_{\text{miss}}^2 \approx 1.4 \times 10^{-17}$. These are prophetic calculations conditional on the measured model; correlated or non-Gaussian drift requires the empirical-tail procedure. A systematic offset of 1 kHz contributes $|\Delta\omega|T = 1.3 \times 10^{-4} \text{ rad}$; an offset of 1 MHz contributes 0.13 rad and must be corrected or included in the guard-band budget. **P2 — species-B schedule.** For a residue channel with $m = 8$ under the P1 parameters, the discrimination bound is $\pi/8$ and $T \leq 23.8/64 \text{ ns} = 0.37 \text{ ns}$; the channel is therefore run at a lower- σ^2 mode or with more frequent commits. **P3 — carry and borrow.** $M = 8$, positional mode: low register $7 + 1 \rightarrow 0$, carry 1; low register $0 - 1 \rightarrow 7$, borrow 1. Modular mode: $7 + 1 \rightarrow 0$, no carry. **P4 — order-finding, mode A.** $N = 15$, $a = 7$: successive products 7, 4, 13, $1 \rightarrow r = 4$ at the fourth commit; repeats at 8, 12. $N = 21$, $a = 2$: 2, 4, 8, 16, 11, $1 \rightarrow r = 6$. Each step is one multiplication by a (three doublings and conditional additions, or a-fold repeated addition)

followed by a commit. **P5 — pair-source benchmark B4.** Preregistered threshold: phase-sum variance $< 0.1 \text{ rad}^2$ over 10^3 trials, single-seat phase uniform (Kuiper test not rejecting uniformity at 5 %), sum tracking of a $\pi/2$ pump-phase step within 0.1 rad.

CLAIMS

1. A winding-number register comprising: a medium element supporting a coherent oscillatory excitation whose phase relative to a reference oscillator is an open phase; an interference stage in which the open phase is combined with at least one other open phase; a commit detector configured to read the combined open phase against the reference oscillator and to resolve it to a nearest whole turn; and an integer store, separate from the interference output, holding the winding number of the register as a persistent integer, the integer store being one of (i) a commit counter incremented or decremented at each committed wrap of the open phase, (ii) a residue register bank of channels having pairwise coprime moduli each holding a residue as a phase, and (iii) a quantized topological state of the medium element.
2. The register of claim 1, wherein the integer store is the commit counter, and the register further comprises a phase comparator referenced to the reference oscillator, a wrap detector registering each crossing of a 2π boundary in either sense during an open-phase stage, and a sampler sampling the open phase at not fewer than two samples per half-turn of relative phase excursion, the commit detector adding the wrap count and the rounded residual open phase to the commit counter at each commit.
3. The register of claim 1, wherein the integer store is the residue register bank, each channel i holding a residue r_i as a phase $r_i \cdot 2\pi/m_i$, the commit detector resolving each channel to a nearest multiple of $2\pi/m_i$, and a recombination unit recovering the integer as $\sum r_i \cdot w_i \bmod M$ with $M = \prod m_i$ and $w_i = (M/m_i) \cdot [(M/m_i)^{-1} \bmod m_i]$.
4. The register of claim 1, wherein the integer store is the quantized topological state and is one of a fluxoid number of a superconducting loop in which a Josephson phase slip is the whole-turn transition, a mode index of a ring resonator, and a period-position of a standing-wave pattern of a two-component condensate.
5. The register of claim 1, further comprising a modulus M being a whole number of turns, a modulus-boundary detector, and a carry/borrow generator configured to emit exactly one carry to a higher-order register when a commit carries the register across its M -th turn upward and exactly one borrow when a commit carries the register across zero downward, the register wrapping to zero or to $M - 1$ respectively.
6. The register of claim 5, further comprising a mode control input selecting between a positional mode in which carry and borrow are enabled and a modular mode in which carry and borrow are disabled and the register performs arithmetic modulo M .
7. A winding arithmetic-logic unit comprising: a plurality of winding-number registers, each comprising a medium element holding an open phase relative to a reference oscillator, an integer

store holding a winding number as a persistent integer separate from the interference output, and a modulus M ; an interference stage combining open phases of selected registers; a commit stage resolving combined open phase to whole turns and updating the integer stores; a carry/borrow element between adjacent registers emitting one quantized whole-turn transition when a commit carries a register across its modulus boundary upward or across zero downward; a mode control input selecting positional or modular arithmetic; and a controller configured to execute addition and subtraction by combine-and-commit and to execute multiplication of two register values by a control sequence of doubling steps of a multiplicand register and conditional modular additions into a product register, each step followed by a commit.

8. The winding arithmetic-logic unit of claim 7, further comprising a radix-conversion coupling between a source register and a destination register of per-turn quanta in an integer ratio q , the controller expressing N turns of the source as $N \cdot q$ turns, or $\lfloor N/q \rfloor$ turns with a remainder, of the destination by combine-and-commit.
9. The winding arithmetic-logic unit of claim 7, wherein at least one register is a residue register bank and the controller converts its residues to a positional register by constant-multiply-and-accumulate using the multiplication path of claim 7.
10. The winding arithmetic-logic unit of claim 7, wherein each register is triplicated, the committed integers are majority-voted, and the reference oscillator for each register is a reference register in the same medium read differentially.
11. A completion-scheduled commit controller for a phase-completion processor, comprising: a stage clock generating a repeating alternation of open-phase interference stages and commit stages; a calibration memory storing a measured phase-drift variance σ^2 , a systematic offset $\Delta\omega$, a guard band δ , and a commit period T satisfying $T \leq \pi^2 / (2\sigma^2 \cdot \ln(C/\epsilon))$ for a computation length C in commits and a target residual error ϵ , and satisfying $|\Delta\omega| \cdot T + \text{tail} \leq \pi - \delta$; a commit trigger configured to cause, at each commit stage, a commit detector to read open phase against a reference oscillator, to resolve it to a nearest whole turn, and to post the result to an integer store; and a comparator applying the guard band δ , values within δ of a decision boundary being re-sampled or flagged rather than resolved.
12. The controller of claim 11, further configured to shorten T adaptively when a runtime witness comprising a boundary-proximity count or a miscount count from redundant registers exceeds a stored threshold.
13. The controller of claim 11, further comprising a validity-state machine having qualify, valid, degraded, invalid, and safe states, the safe state suspending commits, freezing the integer stores read-only, and disabling any pump.
14. A phase-completion processor comprising: a wave-interference medium; a plurality of winding-number registers each having an integer store holding a winding number separate from the interference output; a commit subsystem comprising a reference oscillator and a phase-sensitive commit detector; a controller configured to clock a repeating alternation of open-phase interference stages and irreversible commit stages, each commit stage resolving accumulated phase to whole-

turn values posted to the integer stores; and a host interface for register load and store and commit-triggered readout.

15. A co-signing gate comprising: a parametric pump driven at approximately twice a mode frequency of a medium so as to create correlated mode pairs of wave-vectors k and $-k$ at a first seat and a second seat, the phase sum of the seats being locked to a pump phase; a pump-phase control setting the sense of the shared circulation; a commit detector at each seat referenced to a common reference oscillator; and a correlation monitor configured to compute the variance of the two-seat phase sum over trials and to report the gate valid when the variance lies below a stored threshold while each single-seat phase is distributed uniformly.
16. The co-signing gate of claim 15, packaged as a stand-alone correlation source having mode-select inputs setting a bias field and pump power above a parametric threshold, and providing correlated pairs on demand to a phase-completion processor.
17. A condensate collective register comprising: a ferrimagnetic film; a parametric pump maintaining a magnon condensate in the film at a spectral minimum $\omega_{\min}$; a seed antenna matched to $k_{\min}$ and driven at $\omega_{\min}$ by a signal phase-referenced to a reference oscillator to write the collective phase; a bias-field source configured to apply a field step ΔH for a duration τ to advance the collective phase by $\gamma \Delta H \cdot \tau$; a pickup antenna matched to $k_{\min}$ mixed against the reference oscillator at $\omega_{\min}$; and a commit reader resolving the collective phase to a whole turn after a pump-off delay and posting it to an integer store comprising a commit counter.
18. The condensate collective register of claim 17, wherein the condensate is a two-component condensate of wave-vectors $\pm k_{\min}$ and the integer store is the period-position of its standing-wave density pattern read by spatially resolved Brillouin light scattering.
19. The condensate collective register of claim 17, further comprising a plurality of $k_{\min}$ -matched coupling antennas injecting mode contributions into the condensate for a shared open-phase stage prior to a common commit.
20. A period-finding co-processor comprising: a winding arithmetic-logic unit having a plurality of winding-number registers with integer stores and a modulus, a carry/borrow element, and a controller executing modular addition and modular multiplication by doubling steps and conditional modular additions; a bounded-depth interference stage; a commit readout stage referenced to a reference oscillator; a periodicity extractor identifying a first step r at which a committed register value returns to its initial value; and a host interface receiving a modulus N and a target integer a and returning committed winding values; the controller being operable in at least one of a first mode computing each successive power $a^{\{k+1\}} \bmod N$ from $a^k \bmod N$ by one modular multiplication followed by a commit, and a second mode loading host-supplied values of $a^k \bmod N$ and performing modular accumulation, commit, and periodicity extraction.
21. The period-finding co-processor of claim 20, operated in the second mode, wherein the winding arithmetic-logic unit functions as a modular register memory with committed readout.
22. The period-finding co-processor of claim 20, wherein the interference depth between commits is bounded to the number of combining operations executable within the qualified commit period set

by the controller of claim 11.

23. A method of committing a sum in a winding-number register, comprising: expressing a first addend and a second addend as open phases relative to a reference oscillator in an interference stage; combining the open phases; counting, with a wrap detector, the whole turns crossed by the combined phase during the open-phase stage; at a scheduled commit, reading the combined open phase against the reference oscillator and rounding the residual to a nearest whole turn; and posting to an integer store the sum of the integer parts of the addends, the wrap count, and the rounded residual, the integer store being a commit counter, a residue register bank, or a quantized topological medium state.
24. The method of claim 23, wherein the integer store is a residue register bank, the combining step adds residues modulo m_i in each channel, the rounding step resolves each channel to a nearest multiple of $2\pi/m_i$, and the posting step recombines the residues by Chinese-remainder weighting.
25. A method of scheduling commits in a phase-completion processor, comprising: measuring, with commits disabled, a phase-error distribution and the systematic offset $\Delta\omega$ of a register mode against a reference oscillator; storing the distribution and $\Delta\omega$; selecting a commit period T such that C times a measured or conservatively bounded per-commit whole-turn miscount probability does not exceed a target residual error ϵ for a computation length C , and such that $|\Delta\omega| \cdot T$ plus a stored tail bound does not exceed $\pi - \delta$ for a guard band δ ; clocking open-phase stages of duration not exceeding T alternating with commit stages; and at each commit resolving open phase to a nearest whole turn and posting to an integer store; wherein, when the stored distribution is a qualified zero-mean Gaussian independent-increment model with variance rate σ^2 , T satisfies $T \leq \pi^2 / \{2\sigma^2 [\text{erfc}^{-1}(\epsilon/C)]^2\}$.
26. The method of claim 25, further comprising re-sampling or flagging any open phase lying within δ of a decision boundary at commit, counting such events as a runtime witness, and shortening T when the count exceeds a stored threshold.
27. A method of modular multiplication in winding hardware, comprising: loading a multiplicand into a first winding-number register having a modulus M and an integer store; for each bit of a multiplier from least significant to most significant, doubling the first register by combining its open phase with itself and committing, and, when the bit is one, combining the open phase of the first register with a product register and committing; emitting one carry or one borrow to a higher-order register whenever a commit crosses the modulus boundary in positional mode, or wrapping without carry in modular mode; and reading the product from the integer store of the product register.
28. The method of claim 27, wherein the registers are residue register banks and the doubling and conditional addition are performed residue-wise modulo m_i in each channel.
29. A method of order-finding, comprising: receiving at a host interface a modulus N and a target integer a ; obtaining a sequence of modular powers $a^k \bmod N$ either by computing each power from the preceding power by the method of claim 27 or by loading host-supplied values; expressing each power as an open phase, combining in a bounded-depth interference stage, and

committing to a winding-number register having an integer store; and identifying the order r as the first k at which the committed value returns to its initial value, with continuous-phase precision confined to the bounded stages.

30. The method of claim 29, wherein the powers are host-supplied and the winding hardware performs modular accumulation, commit, and periodicity extraction only.
31. A method of operating a co-signing gate, comprising: driving a parametric pump at approximately twice a mode frequency to create correlated mode pairs at a first seat and a second seat; setting the sense of shared circulation by the pump phase; committing at each seat against a common reference oscillator; and verifying, over repeated trials, that the variance of the two-seat phase sum lies below a stored threshold, that the sum tracks a commanded change of pump phase, and that each single-seat phase is distributed uniformly, before releasing the correlated pairs as a resource.
32. A method of operating a condensate collective register, comprising: maintaining a magnon condensate in a ferrimagnetic film by parametric pumping; writing a collective phase by injecting a seed signal at $\omega_{\min}$ through a $k_{\min}$ -matched antenna phase-referenced to a reference oscillator; steering the collective phase by applying a bias-field step ΔH for a duration τ ; and reading the register by mixing a $k_{\min}$ -matched pickup against the reference oscillator after a pump-off delay and committing the collective phase to a whole turn in a commit counter.
33. A method of qualifying a phase-completion device, comprising: recording open phase against a reference oscillator with commits disabled to fit and store a drift variance σ^2 and an offset $\Delta\omega$; executing a known load-add-commit sequence at commit periods $2T$, T , and $T/2$ and confirming that the miscount rate follows $\text{erfc}(\pi/(\sigma\sqrt{2T}))$ with the stored σ^2 ; recording a boundary-proximity histogram and setting a guard band δ ; confirming over trials that a higher-order register equals $\lfloor \text{total}/M \rfloor$ against a host reference; storing T , δ , M , and mode assignments in a calibration memory with provenance; and entering a valid state only when each acceptance gate passes, otherwise entering an invalid or safe state in which commits are suspended and integer stores are frozen read-only.
34. The co-signing gate of claim 15, further comprising four analyzer-setting inputs and a correlation processor configured to retain every registered trial including loss and no-commit events, to compute four binary-outcome correlations and a CHSH statistic from preregistered settings, and to report whether the magnitude exceeds 2 and approaches $2\sqrt{2}$ within a stored visibility and thermal-occupation envelope, the result being a prophetic discriminator of the two-ended co-signing account rather than a prerequisite for claims 1–14 and 17–33.
35. A method of testing a co-signing gate, comprising: preregistering four analyzer settings, commit windows, validity limits, and treatment of loss and no-commit events; generating paired seat events under a common pump-referenced phase sum; committing a binary result at each seat for every trial; retaining raw singles and joint counts without fair-sampling deletion; computing the four correlations and a CHSH statistic; and comparing the result with both the local classical magnitude 2 and the POAMS-native prophetic value $2\sqrt{2}$.

ABSTRACT

Phase-completion computing devices hold data as the integer winding number of a coherent oscillatory excitation and compute by alternating open-phase interference stages with scheduled irreversible commit stages. A winding-number register comprises a medium element carrying the open phase, a commit detector referenced to a reference oscillator, and an integer store of disclosed species — a commit counter incremented at each committed wrap, a residue register bank recombined by Chinese-remainder weighting, or a topological medium state. A winding arithmetic-logic unit adds, subtracts, and multiplies by doubling and conditional modular addition, with a per-register modulus, carry and borrow at the modulus boundary, and a mode control selecting positional or modular arithmetic. A completion-scheduled commit controller sets the commit period from measured drift variance so sub-turn error is discarded at each commit. Also disclosed: a co-signing parametric pair gate, a condensate collective register, and a period-finding co-processor with hardware or host-supplied modular powers.

PRIOR-ART DISTINCTIONS

Magnonic holographic / interference logic (Khitun; UC Riverside lineage; and Csaba/Porod interference schemes). The prior art performs wave-interference logic and holographic parallel data processing, and proposes prime-factorization by spin-wave interference. It is distinguished in that (a) it provides no integer winding-number registers holding durable, multi-valued state quantized by whole turns in an integer store — its state is phase-amplitude, not a committed integer; (b) it discloses no scheduled irreversible commit stage acting as an error-digitizer between bounded interference stages; and (c) its factoring is of the interference (Gauss-sum) class, subject to ghost-factor precision scaling (the published magnonic-computing literature), whereas the present invention keeps persistent arithmetic in quantized winding space with precision confined to bounded stages.

Parametron and adiabatic quantum-flux-parametron (Goto lineage; AQFP; Kerr-cat/JPO). The prior art encodes a single bit in one of two phase states 180° apart (phase-bistable) and computes by majority logic. It is distinguished in that the present invention (a) uses integer multi-turn winding registers rather than two-state phase bistability, providing multi-valued and modular arithmetic in winding space; (b) implements carry/overflow and borrow as deliberate quantized whole-turn transitions between registers at a modulus boundary under a mode control; and (c) organizes computation as scheduled open-phase/commit alternation with the commit as error-digitizer, rather than continuous phase-locked bistable settling. The parametron is acknowledged as the closest spiritual ancestor and is distinguished on these specific structural grounds.

Oscillator, coherent, and simulated-bifurcation Ising machines (OIM/SHIL; NTT/Stanford CIM; Toshiba SBM). The prior art binarizes coupled-oscillator phase to $0/\pi$ spins and settles to an optimization (Ising ground-state) configuration. It is distinguished in that it provides optimization-only annealing with no arithmetic registers, no winding-number state, no radix/modular arithmetic, and no scheduled commit/period-finding stack; the present invention supplies arithmetic and order-finding, not energy minimization.

Magnon-BEC computing whitespace. Per the patent-landscape record, magnon-condensate *computing* has essentially no issued patent coverage — existing magnon-BEC IP is measurement/spintronic rather than a condensate-phase logic element — and room-temperature phase-coherent wave computing that unifies condensate coherence with interference logic is largely unclaimed (the published patent-landscape record). Family H4 (condensate collective register / interference bus) and its integration with H1–H3, H5 are directed to that whitespace.

Photonic matrix multipliers and analog Fourier/correlator processors. Distinguished as linear-algebra (matrix-vector) or fixed-transform analog engines with no durable quantized state and no scheduled commit arithmetic; cited only as media examples for the alternative-embodiment scope of the alternative-media embodiments.

Rapid single-flux-quantum (RSFQ/SFQ) and related pulse logic. The prior art uses a 2π Josephson phase slip as a pulse primitive and stores integer flux quanta in loops. It is distinguished in that the present invention provides (a) an open-phase interference stage between commits in which continuous phase combines linearly; (b) a scheduled commit acting as an error-digitizer against continuous drift under the relation disclosed in the commit-schedule subsection; (c) multi-radix and residue registers with radix conversion and a mode control between positional and modular arithmetic; and (d) a parametric pair source and condensate bus. The fluxoid-number loop is disclosed herein only as one species-C integer store within that architecture.

Phase-completion processor (1000)

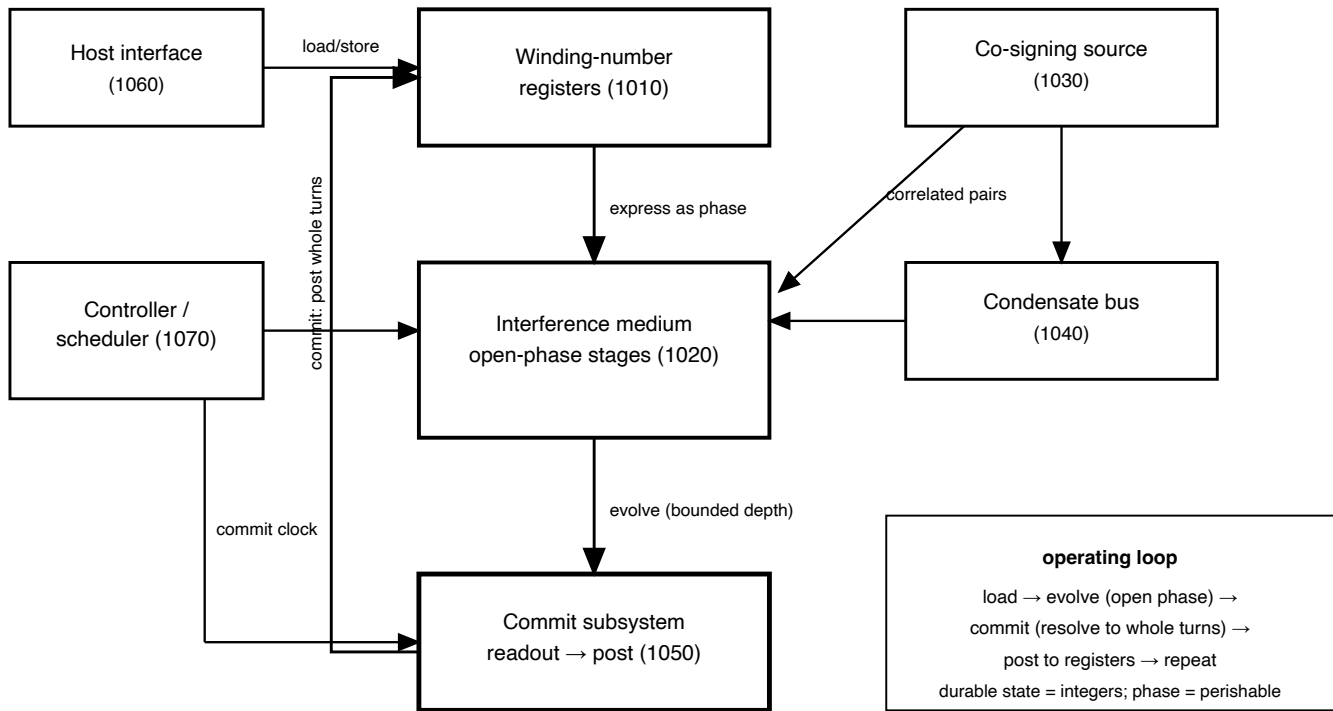


FIG. 1000

System overview — winding-number registers, interference medium, co-signing source, condensate bus, commit subsystem, and host interface; the evolve → commit → post loop.

Winding ALU (2000)

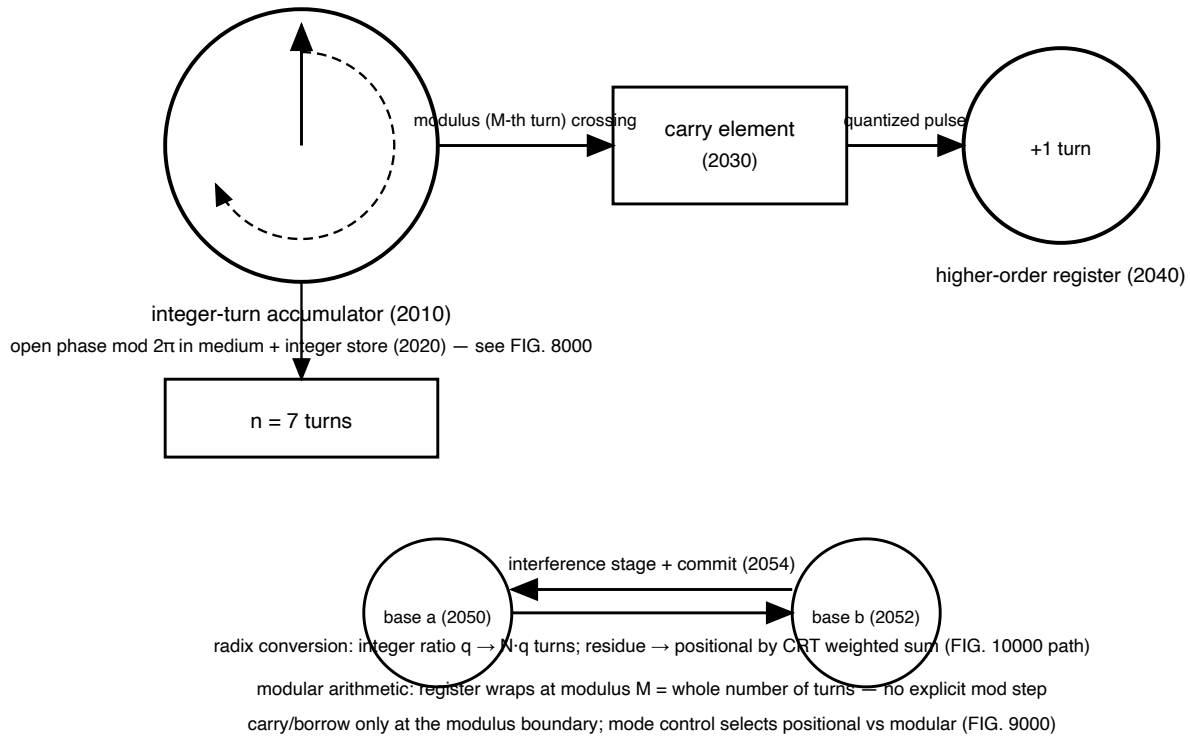


FIG. 2000

H1 winding ALU — integer-turn accumulator with integer store; carry as a quantized whole-turn transition to a higher-order register at the modulus boundary; radix conversion by committing between registers of differing per-turn quanta.

Completion-scheduled operation (3000)

stage clock (3010)

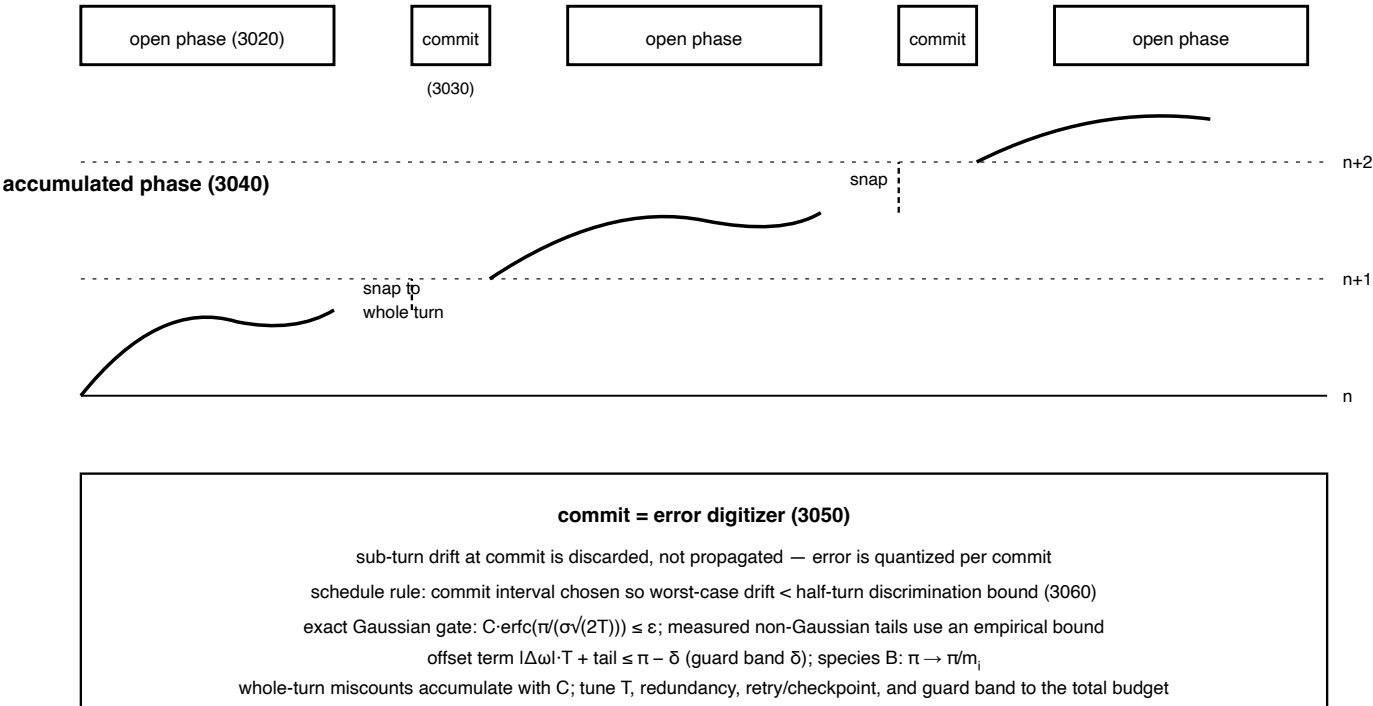
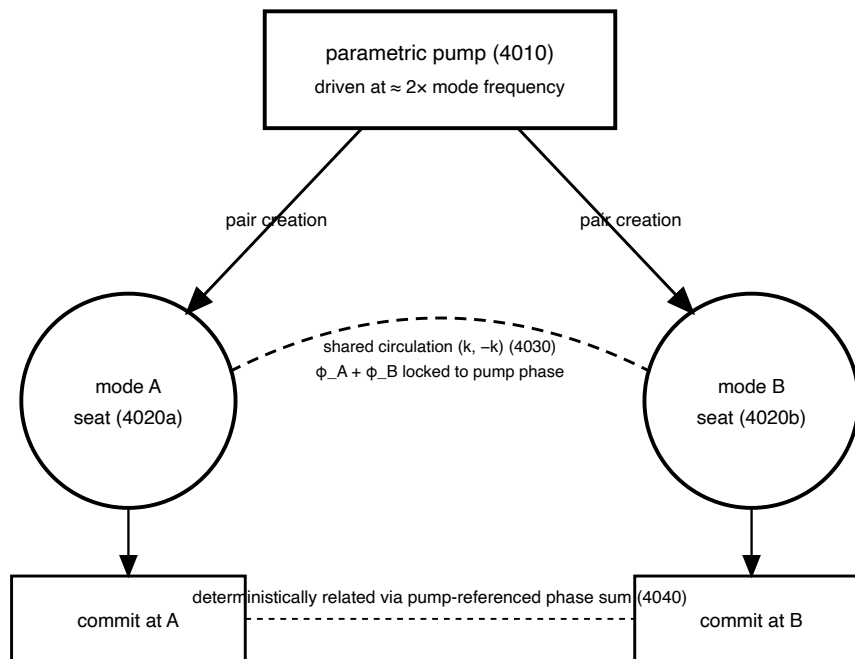


FIG. 3000

H2 completion-scheduled timing — alternation of open-phase and commit stages; accumulated phase snaps to the nearest whole turn at each commit; half-turn discrimination bound and commit-period design relation.

Co-signing gate (4000)



two-seat primitive; stand-alone correlation-source variant (4050): pump + mode select + sense (pump phase) + power > threshold
 commissioning benchmark B4 (classical): $\text{var}(\phi_A + \phi_B)$ below threshold; sum tracks pump phase; single seat uniform;
 commissioning: pump-phase relation · prophetic discriminator: all-trial CHSH approaching $2\sqrt{2}$

FIG. 4000

H3 co-signing gate — parametric pair creation places shared circulation in a defined correlated relation across two seats; commits at the seats are related through the pump-locked phase sum; the classical phase-sum check is a commissioning control, while the prophetic POAMS-native discriminator records all trials and tests a CHSH magnitude approaching $2\sqrt{2}$ without fair-sampling deletion.

Condensate collective register (5000)

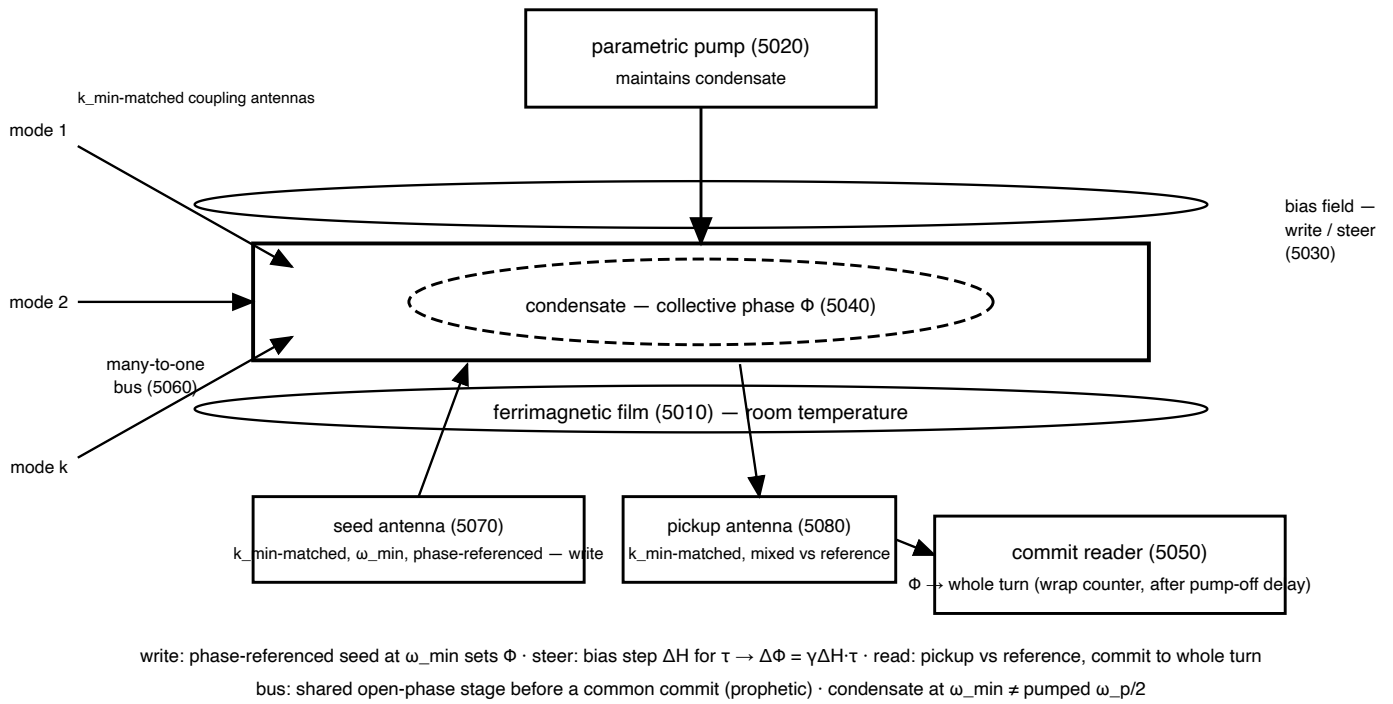


FIG. 5000

H4 condensate collective register — parametrically maintained condensate in a ferrimagnetic film; phase-referenced seed write; bias-field steer; matched-pickup commit read; many-to-one interference bus.

Period-finding engine (6000)

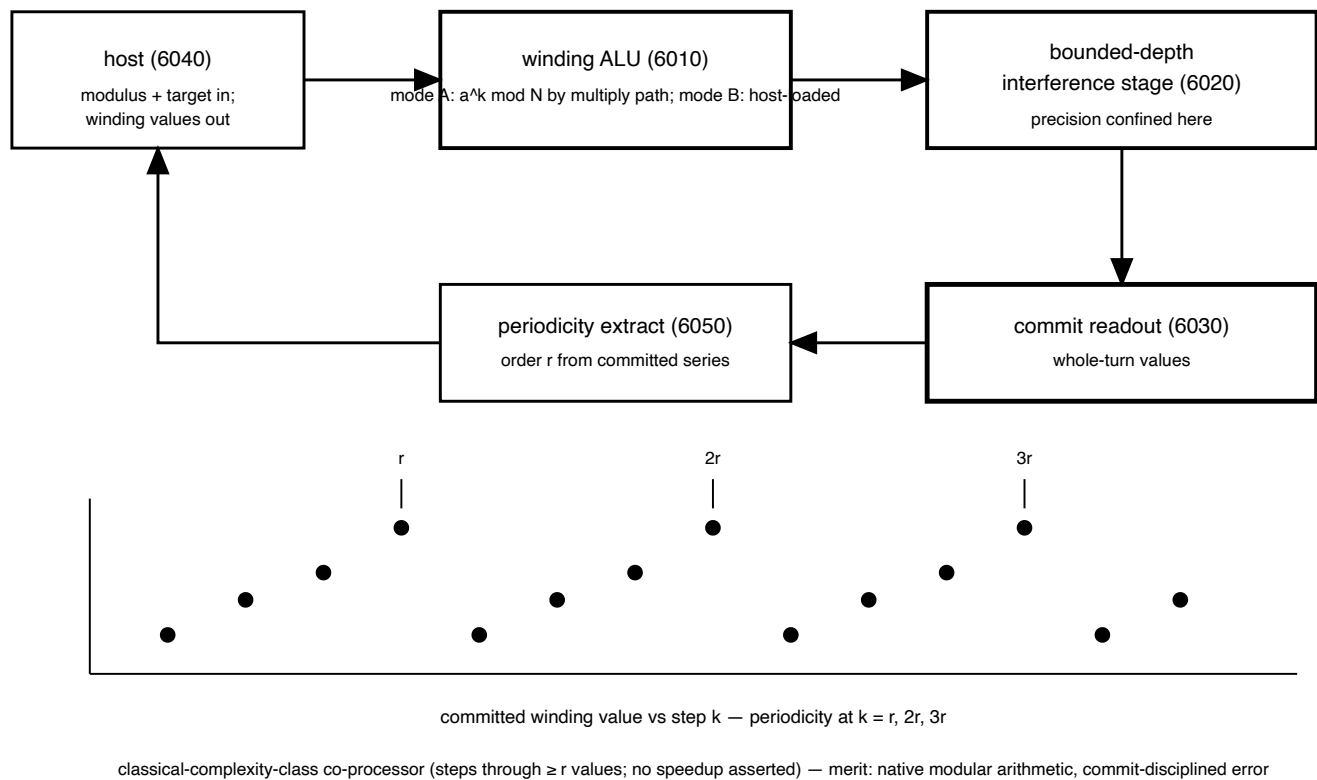
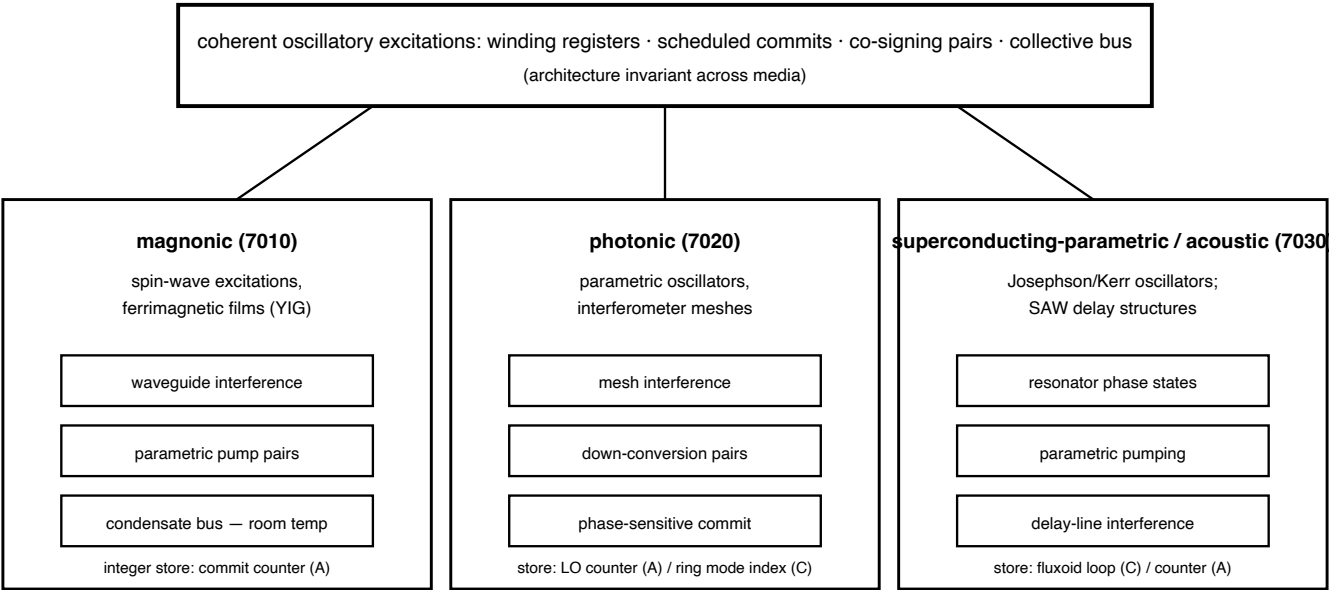


FIG. 6000

H5 period-finding engine — winding ALU (mode A hardware powers / mode B host-supplied powers), bounded-depth interference, and commit readout under a host interface; the order is read from periodicity in the committed winding series.

Alternative media — one architecture (7000)



listing illustrative, not limiting — any medium supporting coherent phase, parametric pairing, and committed readout
winding registers and commit scheduling are medium-independent — alternative media are prophetic embodiments

FIG. 7000

Alternative-media embodiments — the same winding-register / scheduled-commit architecture realized in magnonic, photonic, and superconducting-parametric or acoustic media, each with its named integer-store element.

Integer-store species (8000) — where the integer lives

interference output carries phase mod 2π only; the winding number is held by the integer store and updated at commit

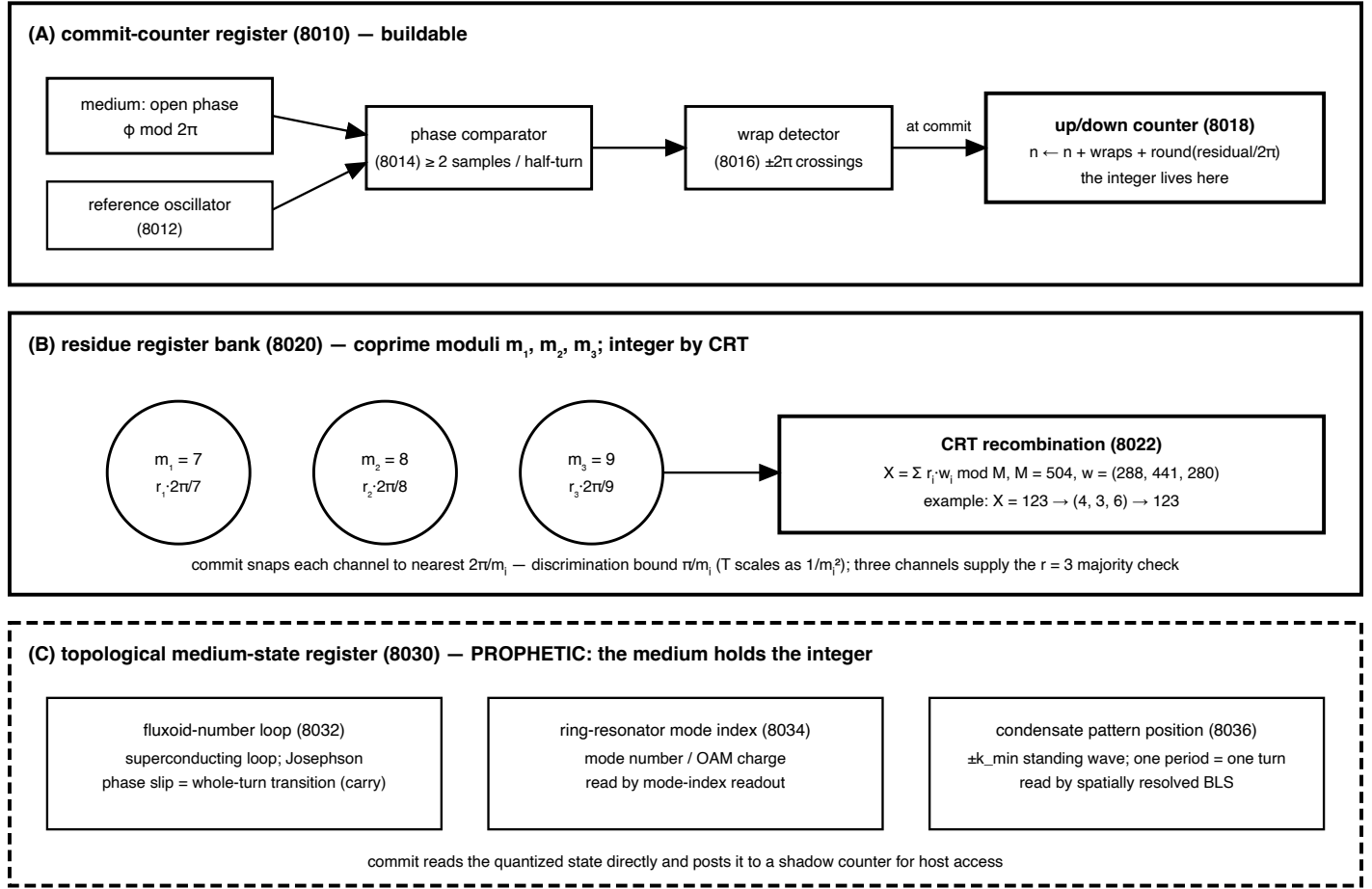


FIG. 8000

Integer-store species — (A) commit-counter register with reference oscillator, phase comparator, wrap detector, and up/down counter; (B) residue register bank with coprime moduli and CRT recombination; (C) topological medium-state register (prophetic).

Modulus, carry, borrow, and mode control (9000)

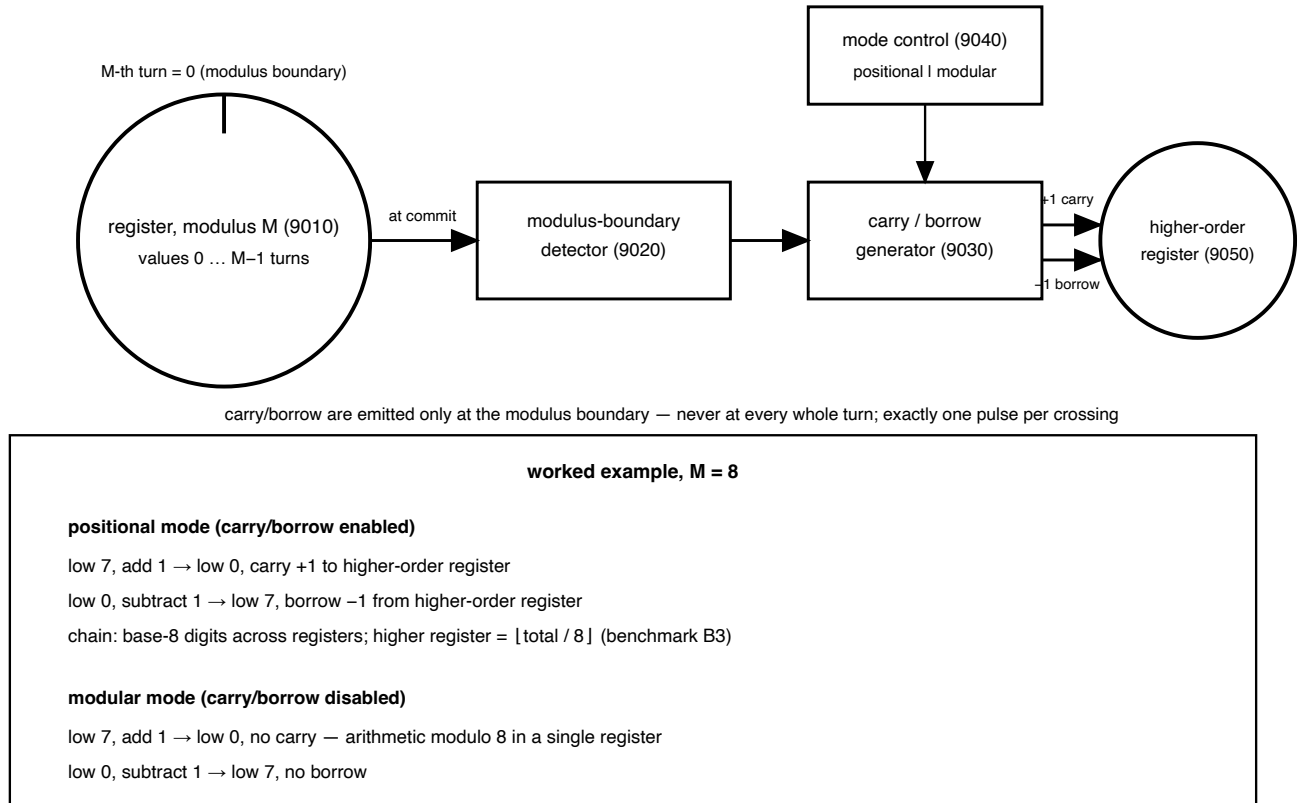
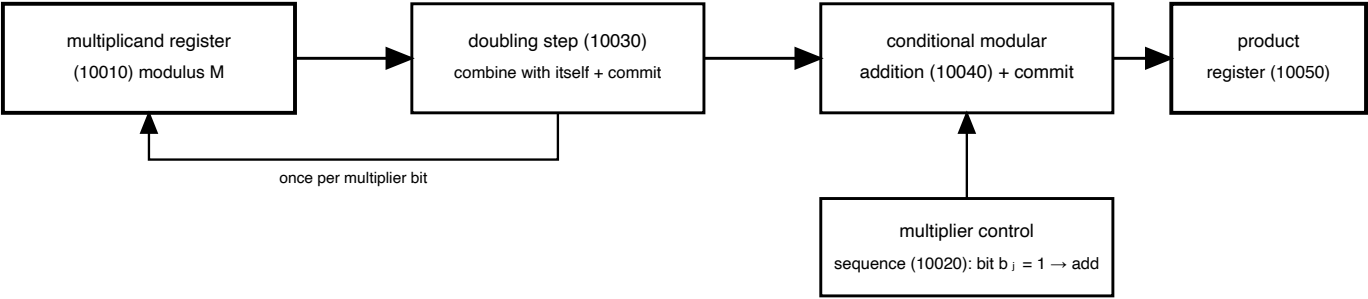


FIG. 9000

Modulus, carry, and borrow control — register with modulus M, modulus-boundary detector, carry/borrow generator, mode control selecting positional or modular arithmetic, higher-order register; worked example M = 8.

Multiplication path and period-finding operating modes (10000)



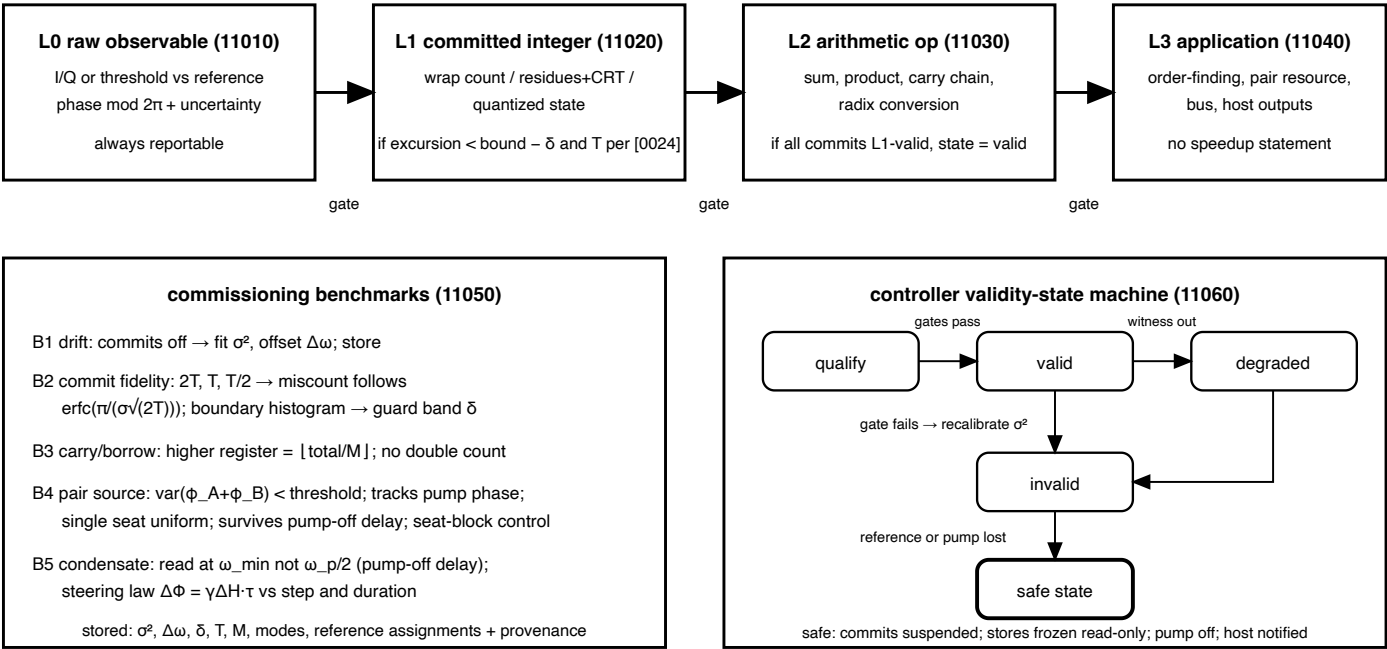
b-bit multiplier: $\leq b$ doublings + b conditional additions, each committed · constant a : a -fold repeated addition · species B: residue-wise mod m_i · harmonic $\times k$ where medium supports

mode A — hardware powers (10060) $a^{\{k+1\}} \bmod N = (a^k \bmod N) \times a$ via the path above one multiplication + commit per step k example $N = 15$, $a = 7$: 7, 4, 13, 1 $\rightarrow r = 4$ example $N = 21$, $a = 2$: 2, 4, 8, 16, 11, 1 $\rightarrow r = 6$ engine = winding ALU + commit + periodicity extractor steps through $\geq r$ values — no asymptotic speedup asserted	mode B — host-supplied powers (10070) host computes $a^k \bmod N$ and loads each value engine: modular accumulate + commit + periodicity extract winding hardware = modular register memory with committed readout and periodicity detector stated honestly: the host performs the multiplication in this mode both modes: continuous-phase precision confined to bounded stages
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FIG. 10000

Multiplication path and engine operating modes — doubling steps and conditional modular additions from the add/wrap/carry primitives; mode A hardware modular exponentiation by iterated multiplication; mode B host-supplied modular powers.

Layered scope and qualification (11000)



runtime witnesses: reference-register phase · boundary-proximity count · miscount count from $r = 3$ copies

degraded $\rightarrow$ shorten T; invalid $\rightarrow$ recalibrate; a deployed unit never reports an L2 result its qualification does not support

prophetic example P1: $\sigma^2 = 0.01 \text{ rad}^2/\text{ns}$, $C = 10^6$, $\epsilon = 10^{-3} \rightarrow T \leq 23.8 \text{ ns}$; $T = 20 \text{ ns} \rightarrow p_{\text{miss}} \approx 2.1 \times 10^{12}$, $3C \cdot p_{\text{miss}}^2 \approx 1.4 \times 10^{17}$

(σ^2 to be measured in B1; no value represented as achieved)

FIG. 11000

Layered scope and qualification — L0 raw interference observable to L3 application with validity gates; commissioning benchmarks B1–B5; controller validity-state machine with safe state.